

Ch 7 Transcendental Functions

Most of the functions (polynomial, rational, radical, etc.) we have studied thus far have been algebraic functions. Algebraic functions involve basic operations, powers, and roots. In this chapter, we discuss exponential functions and logarithmic functions. Exponential functions model a wide variety of phenomena of interest in science, engineering, mathematics, and economics. We use the methods of calculus to obtain rigorous and precise definitions and properties of the exponential functions. We first define the natural logarithm function $y = \ln x$ as a certain integral, and then the natural exponential function $y = e^x$ as its inverse function. These two foundational functions form the basis for the full array of logarithmic and exponential functions. We also introduce inverse trigonometric functions, as well as hyperbolic functions and their inverses, and investigate their applications. Along with the trigonometric functions, all of these functions belong to the category of *transcendental functions*.

7.1 Inverse Functions and Their Derivatives

A function that undoes, or inverts, the effect of a function f is called the inverse of f . Many common functions, though not all, are paired with an inverse. Important inverse functions often show up in applications. Inverse functions also play a key role in the development and properties of the exponential functions. To have an inverse, a function must possess a special property over its domain.

① One-to-One Functions

◆ A function is a rule that assigns each element in its domain to only one value from its range. Some functions assign more than one element in the domain to the same range value. For example, the function $f(x) = x^2$ assigns both of the numbers -1 and $+1$ to 1 .

DEFINITION A function $f(x)$ is **one-to-one** on a domain D if $f(x_1) \neq f(x_2)$

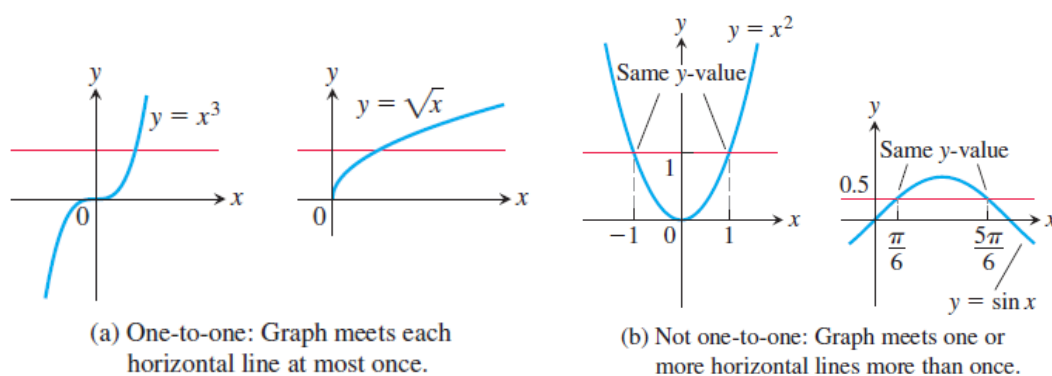
whenever $x_1 \neq x_2$ in D .

Example 1 $f(x) = \sqrt{x}$ is one-to-one on any domain of nonnegative numbers because $\sqrt{x_1} \neq \sqrt{x_2}$ whenever $x_1 \neq x_2$.

Example 2 $f(x) = x^2$ is not one-to-one on $[-2, 2]$ since $1 \neq -1$, but $f(1) = f(-1)$.

◆ The Horizontal Line Test for One-to-One Functions

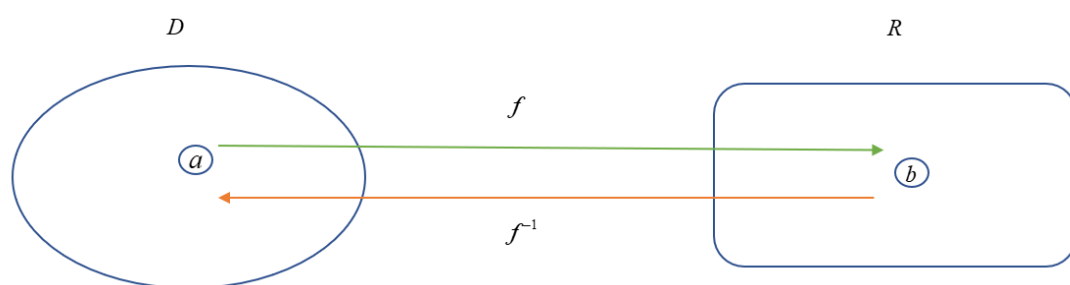
A function $y = f(x)$ is **one-to-one** if and only if its graph intersects each horizontal line at most once.



② Inverse Functions

DEFINITION Suppose that f is a one-to-one function on a domain D with range R .

The **inverse function** f^{-1} is defined by $f^{-1}(b) = a$ if $f(a) = b$.



The domain of f^{-1} is R and the range of f^{-1} is D .

③ Finding Inverses

◆ The process of passing from f to f^{-1} can be summarized as a two-step procedure.

1. Solve the equation $y = f(x)$ for x . This gives a formula $x = f^{-1}(y)$ where x is expressed as a function of y .
2. Interchange x and y , obtaining a formula $y = f^{-1}(x)$ where f^{-1} is expressed in the conventional format with x as the independent variable and y as the dependent variable.

Example 3 Find the inverse of $y = \frac{1}{2}x + 1$, expressed as a function of x .

Example 4 Find the inverse of the function $y = x^2$, $x \geq 0$, expressed as a function of x .

④ Derivatives of Inverses of Differentiable Functions

THEOREM 1—The Derivative Rule for Inverses

If f has an interval I as domain and $f^{-1}(x)$ exists and is never zero on I , then f^{-1} is differentiable at every point in its domain (the range of f). The value of $(f^{-1})'$ at a point b in the domain of f^{-1} is the reciprocal of the value of f' at the point $a = f^{-1}(b)$:

$$(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))} = \frac{1}{f'(a)}$$

or

$$\left. \frac{df^{-1}}{dx} \right|_{x=b} = \frac{1}{\left. \frac{df}{dx} \right|_{x=f^{-1}(b)}}.$$

Example 5 Let $f(x) = x^3 - 2$. Find $(f^{-1})'(6)$.

Example 6 Let $f(x) = x^2 - 4x - 5$, $x > 2$. Find $(f^{-1})'(0)$.